UK Maths Trust

Primary Kangaroo

Thursday 20 March – Thursday 27 March 2025

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England & Wales: Year 6 or below | *Scotland: P7 or below* | *Northern Ireland: Year 7 or below*

Instructions

1. Do not open the paper until the invigilator tells you to do so.
2. Time allowed: **45 minutes**.
   No answers, or personal details, may be entered after the allowed time is over.
3. The use of blank or lined paper for rough working is allowed; **squared paper**, **calculators and measuring instruments are forbidden**.
4. **Use a B or an HB non-propelling pencil.** Mark at most one of the options A, B, C, D, E on the Answer Sheet for each question. Do not mark more than one option.
5. **Do not expect to finish the whole paper in the time allowed.** The questions in this paper have been arranged in approximate order of difficulty with the harder questions towards the end. You are not expected to complete all the questions during the time. You should bear this in mind when deciding which questions to tackle.
6. **Scoring rules:**
   5 marks are awarded for each correct answer.
   You will not lose marks for getting answers wrong.
7. **Your Answer Sheet will be read by a machine.** Do not write or doodle on the sheet except to mark your chosen options. The machine will read all black pencil markings even if they are in the wrong places. If you mark the sheet in the wrong place, or leave bits of eraser stuck to the page, the machine will interpret the mark in its own way, or reject the answer sheet.
8. **The questions on this paper are designed to challenge you to think, not to guess.** You will gain more marks, and more satisfaction, by doing one question carefully than by guessing lots of answers. This paper is about solving interesting problems, not about lucky guessing.

Enquiries about the Primary Kangaroo should be sent to:

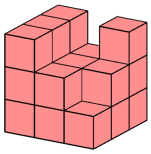
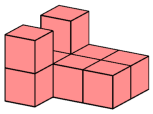
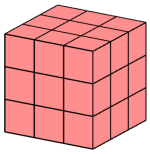
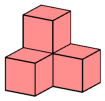
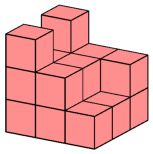
challenges@ukmt.org.uk

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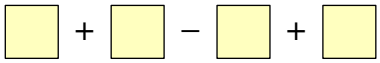
supported by Overleaf

1. Mia is adding small cubes one by one to build a $3 \times 3 \times 3$ cube.
She took the five photographs shown below at five different times.

What does the fourth photograph that Mia took look like?

A  B  C  D  E 

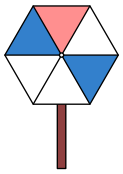
2. Simon writes the four digits 2, 0, 2, 5 in the four boxes below.



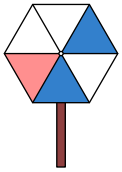
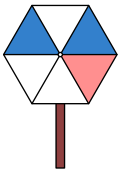
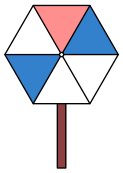
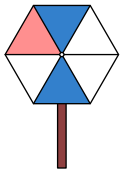
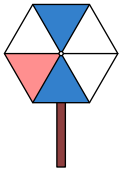
Which order would give him the largest result?

A 0, 2, 2, 5 B 0, 5, 2, 2 C 2, 5, 2, 0 D 5, 0, 2, 2 E 5, 2, 0, 2

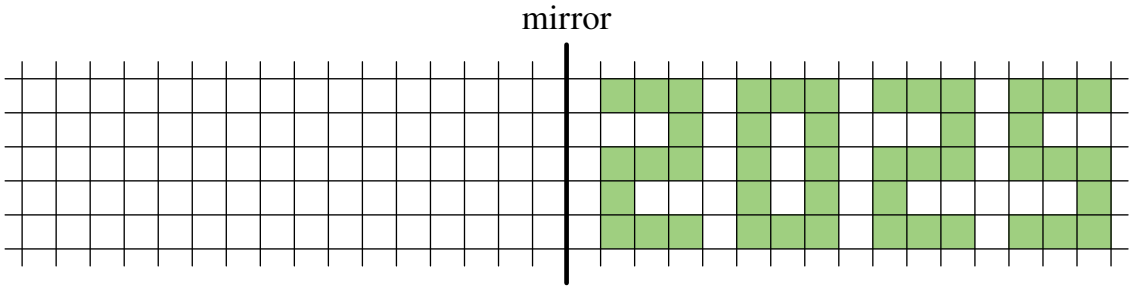
3. Larissa has a pinwheel, which is shown below.



She spins her pinwheel. Which of the pinwheels shown below could be hers?

A  B  C  D  E 

4. Otto reflects the number 2025 in the mirror on its left.



What is the difference between 2025 and its reflection?

A 350 B 477 C 480 D 2840 E 2997

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5. On a normal dice, the total number of spots on two opposite faces is always 7.

Only one of the dice shown below could be a normal dice. Which one?

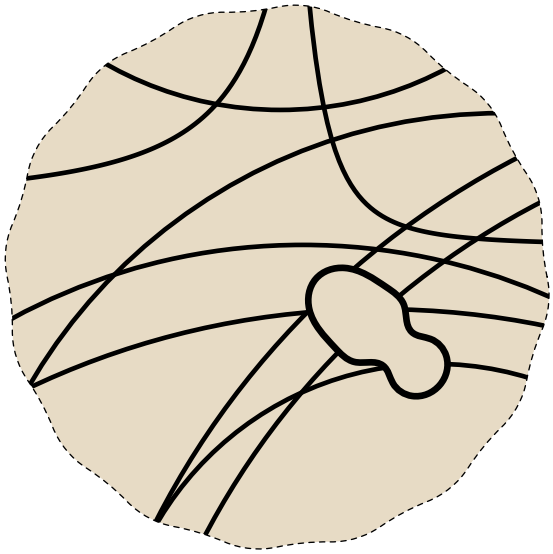
A  B  C  D  E 

6. Alice stepped on some tracks on the ground.

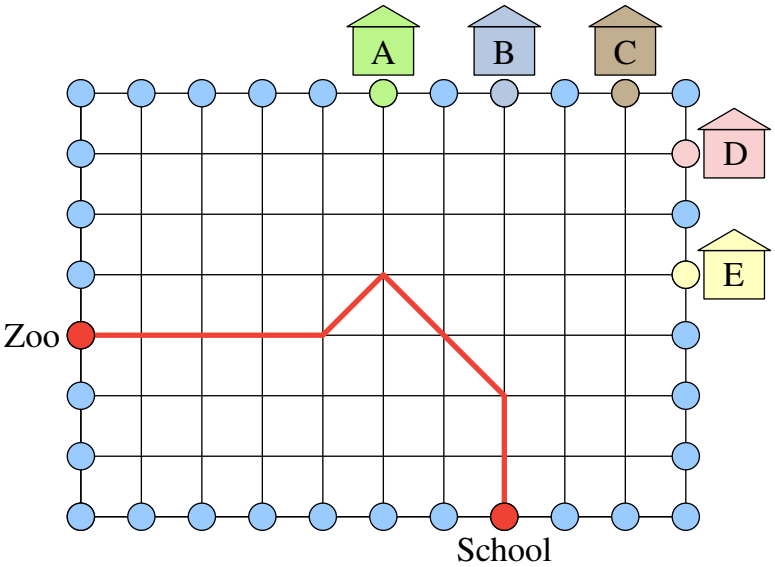
Which track pattern was under her foot?

A  B  C 

D  E 



7. Kangaroo jumps from School to the Zoo, as follows: ↑ 2, ↖ 2, ↙ 1, ← 4, as shown below:



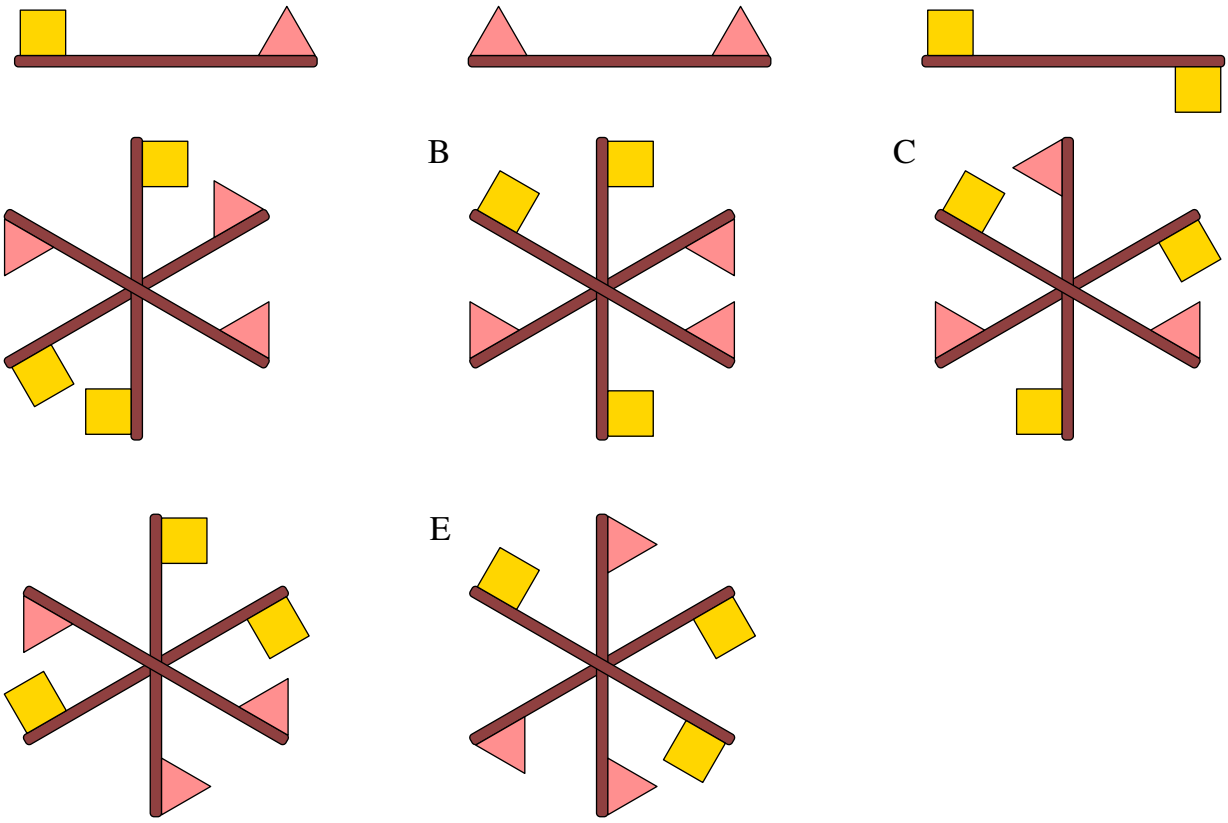
Now he jumps from the Zoo, as follows: → 3, ↗ 2, ↑ 2.

Which house will he arrive at?

A  B  C  D  E 

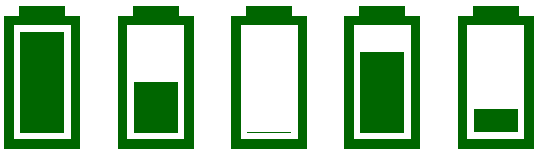
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8. Which pin-wheel can Jorge build with these three rods?



9. In the morning, five friends (Ann, Bob, Cristina, David and Edward) had the same 'fully-charged' mobile phones. By the evening, Bob had spoken on the phone as much as Ann and Cristina together. Bob's phone ran out of power. David had not used his phone at all.

The battery levels of their phones are shown on the right.

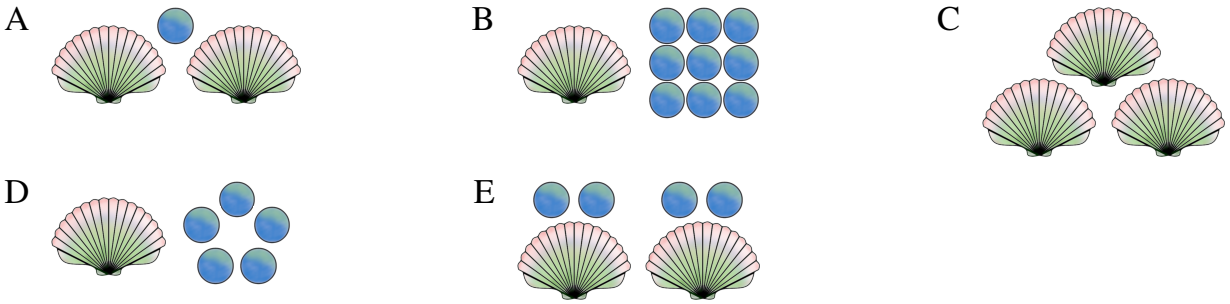


Which phone belongs to Edward?

A 1 B 2 C 3 D 4 E 5

10. Nico and his little sister are playing with shells and marbles.
In their shop, each shell has a value of 6 and each marble has a value of 1.

Which of the following collection of shells and marbles has a total value of 16?

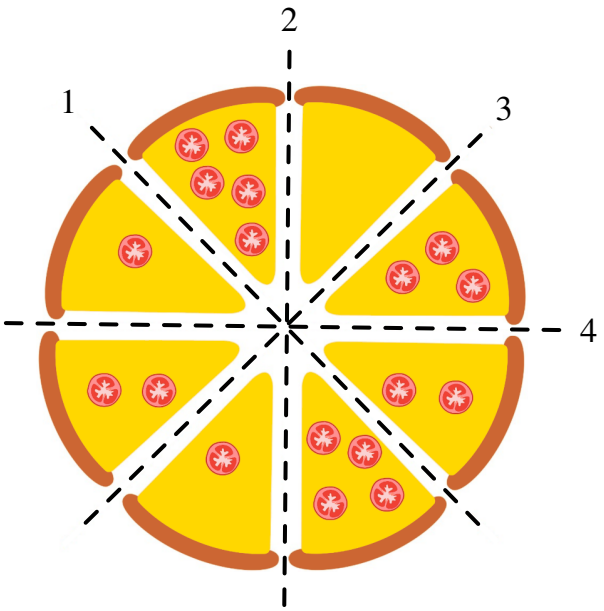


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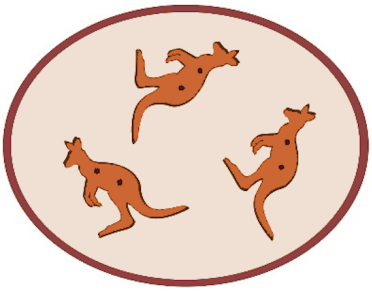
11. Tom wishes to slice a pizza into 2 halves. He also wishes to have the same number of tomatoes on each half. It is possible for him to do this with two different cuts.

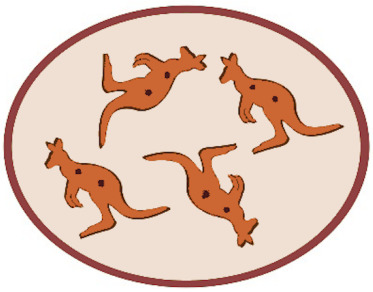
Along which lines could he cut?

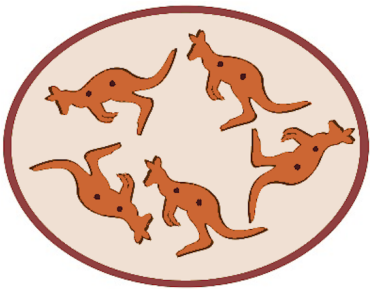
A 1 and 3 B 1 and 4 C 2 and 3
D 2 and 4 E 3 and 4



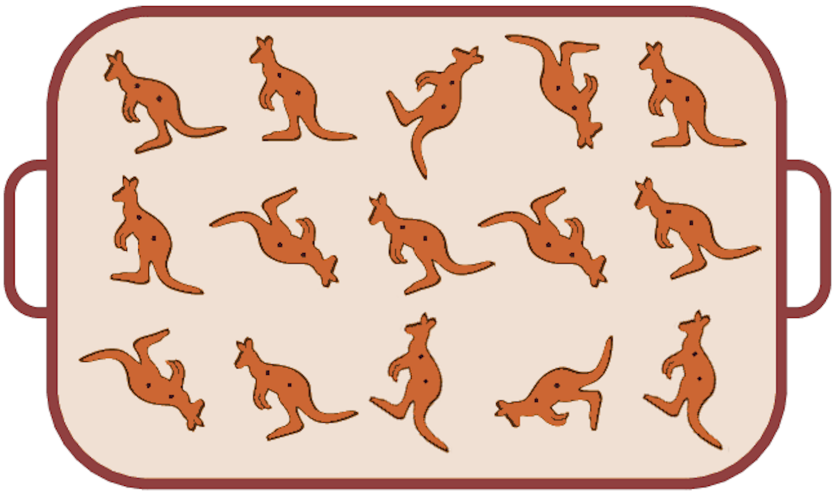
12. Anna, Bonnie and Caspar already have some kangaroo cookies on their plates.


Anna


Bonnie


Caspar

They share the remaining 15 cookies on the tray shown below so that everyone will have the **same** number of cookies on their plates.

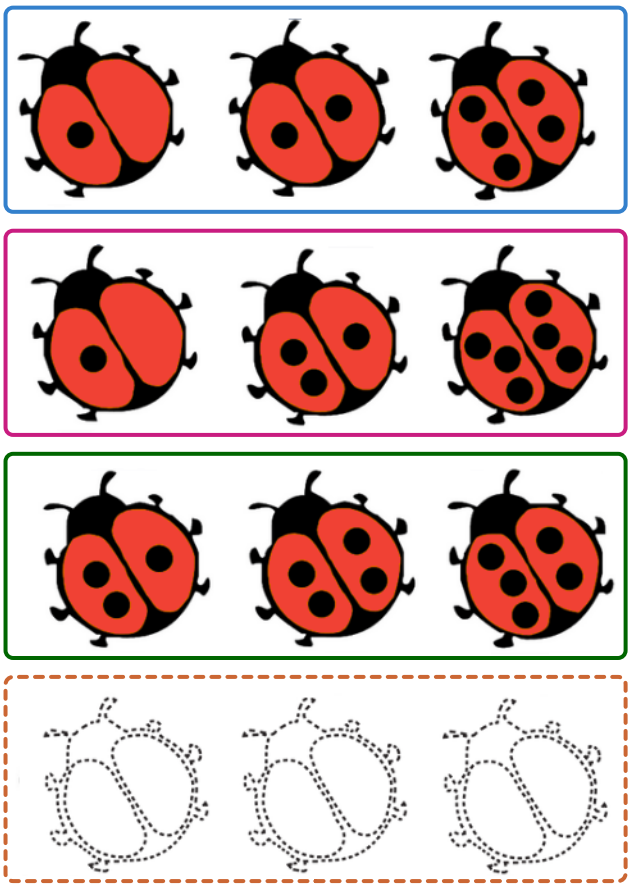


How many *more* cookies must Anna receive?

A 4 B 5 C 6 D 7 E 8

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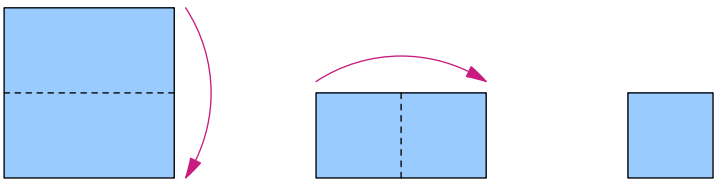
13. Six ladybirds have 1, 2, 3, 4, 5 or 6 spots each. Marta took four photos of them in groups of 3. Each ladybird appeared the same number of times in the four photos. Three of the photos are shown here, along with the outline of the fourth photo.



How many spots do the three ladybirds in Marta's fourth photo have in total?

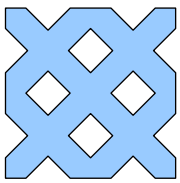
A 9 B 10 C 11 D 12 E 23

14. Nela folds a paper square in half and then in half again, as shown.



Next she cuts pieces out of the folded paper.

After unfolding, she sees a paper snowflake.

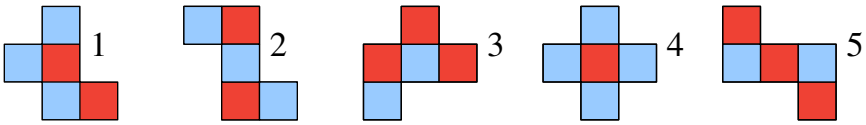


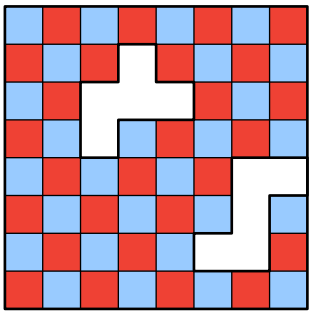
How did she cut the folded paper?

A 
B 
C 
D 
E 

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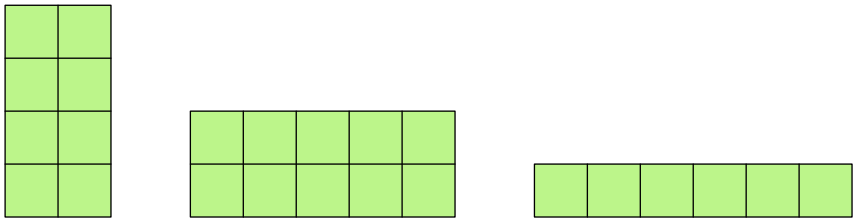
15. Which two of the pieces below complete the chessboard?



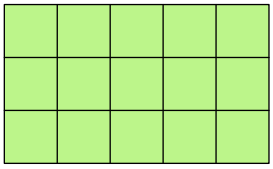


A pieces 1 and 2 B pieces 1 and 5 C pieces 3 and 4
D pieces 3 and 5 E pieces 4 and 5

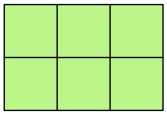
16. Bob makes a square from four rectangular pieces. Three of the pieces are shown below.

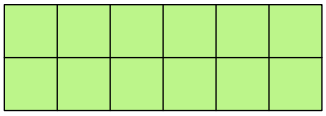


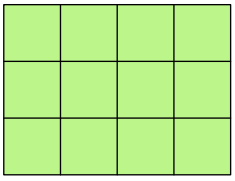
Which one is the fourth piece?

A 

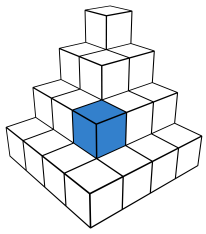
B 

C 

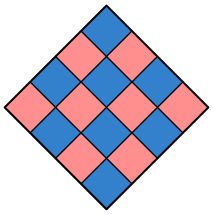
D 

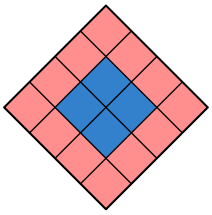
E 

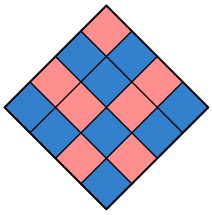
17. Léonie stacks up some pink and blue cubes into a pyramid as shown on the right.
One of the blue cubes is shown.
She arranges each cube so that each of its faces does **not** touch a face of another cube that has the same colour.

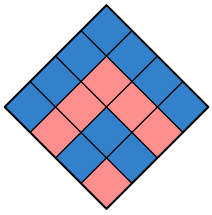


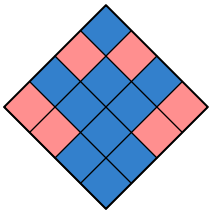
What will Léonie's pyramid look like from above?

A 

B 

C 

D 

E 

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18. The picture shows the page for one month of a calendar without any of the dates.

The total of the dates for the two shaded days is 29.

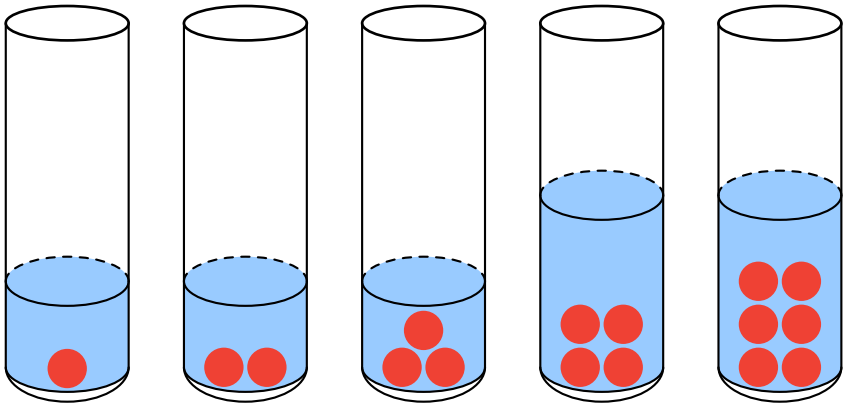
On what day of the week does the first day of this month fall?

Mon	Tue	Wed	Thu	Fri	Sat	Sun

A Monday B Tuesday C Wednesday D Thursday E Sunday

19. Identical balls were placed in five identical test tubes as shown.

Then water was added to each of these test tubes.

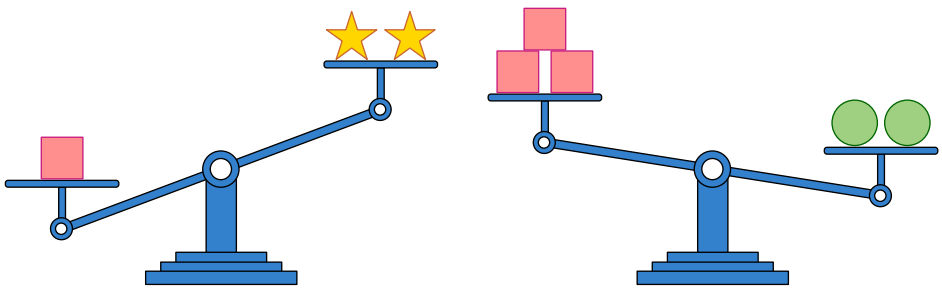


The water level in test tubes 1, 2, and 3 is the same.
The water level in test tubes 4 and 5 is also the same but is twice as high as in the first three.

When all the balls are removed, which test tube will contain the *least* amount of water?

A test tube 1 B test tube 2 C test tube 3 D test tube 4 E test tube 5

20. A pair of scales is used to weigh 3 different objects, and the results are shown below.



Each kind of object (cube, star or ball) has a different mass. The masses can be 1, 2, 3, 4 or 5 kg.

What is the mass in kilograms of one cube?

A 1 B 2 C 3 D 4 E 5

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